Experiment 6: Implementation of Various Join Operations

Aim: *Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.*

1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

SELECT m.MatchID, m.Ground, m.Date, p.FName, p.LName FROM ODIMatch m INNER JOIN Batting b ON m.MatchID = b.MatchID INNER JOIN Player p ON b.PID = p.PlayerID;

```
MySQL  localhost:33060+ ssl  SQL > USE aarya61_match;
Default schema set to `aarya61_match`.
Fetching global names, object names from `aarya61_match` for auto-completion... Press ^C to stop.
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT m.MatchID, m.Ground, m.Date, p.FName, p.LName
                                               -> FROM ODIMatch m
                                               -> INNER JOIN Batting b ON m.MatchID = b.MatchID
                                               -> INNER JOIN Player p ON b.PID = p.PlayerID;
+---------+----------+------------+-----------+-------------+
| MatchID | Ground   | Date       | FName     | LName       |
+---------+----------+------------+-----------+-------------+
|    2755 | Colombo  | 2008-08-24 | Yuvraj    | Singh       |
|    2755 | Colombo  | 2008-08-24 | MS        | Dhoni       |
|    2755 | Colombo  | 2008-08-24 | Sanath    | Jayasuriya  |
|    2755 | Colombo  | 2008-08-24 | Chaminda  | Vaas        |
|    2755 | Colombo  | 2008-08-24 | Muthiah   | Murlitharan |
|    2689 | Brisbane | 2008-03-04 | Sachin    | Tendulkar   |
|    2689 | Brisbane | 2008-03-04 | Yuvraj    | Singh       |
|    2689 | Brisbane | 2008-03-04 | MS        | Dhoni       |
|    2689 | Brisbane | 2008-03-04 | Adam      | Gilchrist   |
|    2689 | Brisbane | 2008-03-04 | Ricky     | Ponting     |
|    2689 | Brisbane | 2008-03-04 | Andrew    | Symonds     |
|    2689 | Brisbane | 2008-03-04 | Breet     | Lee         |
|    2689 | Brisbane | 2008-03-04 | Praveen   | Kumar       |
|    2755 | Colombo  | 2008-08-24 | Praveen   | Kumar       |
|    2689 | Brisbane | 2008-03-04 | Harbhajan | Singh       |
|    2755 | Colombo  | 2008-08-24 | Harbhajan | Singh       |
+---------+----------+------------+-----------+-------------+
16 rows in set (0.0349 sec)
```

2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

SELECT CONCAT(p.FName, ' ', p.LName) AS PlayerName, p.Country, b.NRuns, b.Six FROM Player p INNER JOIN Batting b ON p.PlayerID = b.PID WHERE b.Six > 0;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT CONCAT(p.FName, ' ', p.LName) AS PlayerName,
                                               ->        p.Country, b.NRuns, b.Six
                                               -> FROM Player p
                                               -> INNER JOIN Batting b ON p.PlayerID = b.PID
                                               -> WHERE b.Six > 0;
+-------------------+-----------+-------+-----+
| PlayerName        | Country   | NRuns | Six |
+-------------------+-----------+-------+-----+
| Sanath Jayasuriya | Sri Lanka |    60 |   2 |
| Yuvraj Singh      | India     |    38 |   2 |
| MS Dhoni          | India     |    36 |   1 |
| Andrew Symonds    | Australia |    42 |   1 |
+-------------------+-----------+-------+-----+
4 rows in set (0.0015 sec)
```

3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

SELECT p.PlayerID, p.FName, p.LName, b.MatchID, b.NRuns FROM Player p LEFT OUTER JOIN Batting b ON p.PlayerID = b.PID;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT p.PlayerID, p.FName, p.LName, b.MatchID, b.NRuns
                                                 -> FROM Player p
                                                 -> LEFT OUTER JOIN Batting b ON p.PlayerID = b.PID;
+----------+------------+-------------+---------+-------+
| PlayerID | FName      | LName       | MatchID | NRuns |
+----------+------------+-------------+---------+-------+
|    21001 | Virender   | Sehwag      |    NULL |  NULL |
|    23001 | Yuvraj     | Singh       |    2755 |     0 |
|    23001 | Yuvraj     | Singh       |    2689 |    38 |
|    24001 | Andrew     | Symonds     |    2689 |    42 |
|    25001 | MS         | Dhoni       |    2755 |    71 |
|    25001 | MS         | Dhoni       |    2689 |    36 |
|    27001 | Praveen    | Kumar       |    2689 |     7 |
|    27001 | Praveen    | Kumar       |    2755 |     2 |
|    27002 | Ishant     | Sharma      |    NULL |  NULL |
|    89001 | Sachin     | Tendulkar   |    2689 |    91 |
|    90001 | Brain      | Lara        |    NULL |  NULL |
|    90002 | Anil       | Kumble      |    NULL |  NULL |
|    91001 | Sanath     | Jayasuriya  |    2755 |    60 |
|    92001 | Shane      | Warne       |    NULL |  NULL |
|    92002 | Muthiah    | Murlitharan |    2755 |     1 |
|    92003 | Inzamam    | UI-Huq      |    NULL |  NULL |
|    93002 | Heath      | Streak      |    NULL |  NULL |
|    93003 | Gary       | Kirsten     |    NULL |  NULL |
|    94002 | Chaminda   | Vaas        |    2755 |    17 |
|    94004 | Stephen    | Fleming     |    NULL |  NULL |
|    95001 | Ricky      | Ponting     |    2689 |     1 |
|    95002 | Shaun      | Pollock     |    NULL |  NULL |
|    95003 | Jacques    | Kallis      |    NULL |  NULL |
|    96001 | Rahul      | Dravid      |    NULL |  NULL |
|    96002 | Herschelle | Gibbs       |    NULL |  NULL |
|    96003 | Saurav     | Ganguly     |    NULL |  NULL |
|    97004 | Daniel     | Vettori     |    NULL |  NULL |
|    98001 | Shahid     | Afridi      |    NULL |  NULL |
|    98002 | Harbhajan  | Singh       |    2689 |     3 |
|    98002 | Harbhajan  | Singh       |    2755 |     2 |
|    99001 | Breet      | Lee         |    2689 |     7 |
|    99002 | Adam       | Gilchrist   |    2689 |     2 |
|    99003 | Michael    | Vaughan     |    NULL |  NULL |
|   300311 | Michael    | Clark       |    NULL |  NULL |
|   406317 | Sam        | NULL        |    NULL |  NULL |
+----------+------------+-------------+---------+-------+
35 rows in set (0.0045 sec)
```

4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

SELECT bw.MatchID, bw.NOvers, bw.Nwickets, p.FName, p.LName, p.Country FROM Player p RIGHT OUTER JOIN Bowling bw ON p.PlayerID = bw.PID;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT bw.MatchID, bw.NOvers, bw.Nwickets,
                                                 ->        p.FName, p.LName, p.Country
                                                 -> FROM Player p
                                                 -> RIGHT OUTER JOIN Bowling bw ON p.PlayerID = bw.PID;
+---------+--------+----------+-----------+-------------+-----------+
| MatchID | NOvers | Nwickets | FName     | LName       | Country   |
+---------+--------+----------+-----------+-------------+-----------+
|    2689 |     10 |        1 | Breet     | Lee         | Australia |
|    2689 |      3 |        1 | Andrew    | Symonds     | Australia |
|    2689 |      3 |        0 | Yuvraj    | Singh       | India     |
|    2755 |      9 |        1 | Chaminda  | Vaas        | Sri Lanka |
|    2755 |     10 |        1 | Muthiah   | Murlitharan | Sri Lanka |
|    2755 |      4 |        0 | Sanath    | Jayasuriya  | Sri Lanka |
|    2755 |     10 |        2 | Yuvraj    | Singh       | India     |
|    2755 |     10 |        3 | Harbhajan | Singh       | India     |
|    2689 |     10 |        1 | Harbhajan | Singh       | India     |
+---------+--------+----------+-----------+-------------+-----------+
9 rows in set (0.0013 sec)
```

5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

SELECT COALESCE(b.MatchID, bw.MatchID) AS MatchID, COALESCE(b.PID, bw.PID) AS PID, b.NRuns, bw.NOvers, bw.Nwickets

FROM Batting b LEFT JOIN Bowling bw ON b.MatchID = bw.MatchID AND b.PID = bw.PID

UNION

SELECT COALESCE(b.MatchID, bw.MatchID) AS MatchID, COALESCE(b.PID, bw.PID) AS PID, b.NRuns, bw.NOvers, bw.Nwickets

FROM Batting b RIGHT JOIN Bowling bw ON b.MatchID = bw.MatchID AND b.PID = bw.PID;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT COALESCE(b.MatchID, bw.MatchID) AS MatchID,
                                                 ->        COALESCE(b.PID, bw.PID) AS PID,
                                                 ->        b.NRuns, bw.NOvers, bw.Nwickets
                                                 -> FROM Batting b
                                                 -> LEFT JOIN Bowling bw ON b.MatchID = bw.MatchID AND b.PID = bw.PID
                                                 -> UNION
                                                 -> SELECT COALESCE(b.MatchID, bw.MatchID) AS MatchID,
                                                 ->        COALESCE(b.PID, bw.PID) AS PID,
                                                 ->        b.NRuns, bw.NOvers, bw.Nwickets
                                                 -> FROM Batting b
                                                 -> RIGHT JOIN Bowling bw ON b.MatchID = bw.MatchID AND b.PID = bw.PID;
+---------+-------+-------+--------+----------+
| MatchID | PID   | NRuns | NOvers | Nwickets |
+---------+-------+-------+--------+----------+
|    2755 | 23001 |     0 |     10 |        2 |
|    2755 | 25001 |    71 |   NULL |     NULL |
|    2755 | 91001 |    60 |      4 |        0 |
|    2755 | 94002 |    17 |      9 |        1 |
|    2755 | 92002 |     1 |     10 |        1 |
|    2689 | 89001 |    91 |   NULL |     NULL |
|    2689 | 23001 |    38 |      3 |        0 |
|    2689 | 25001 |    36 |   NULL |     NULL |
|    2689 | 99002 |     2 |   NULL |     NULL |
|    2689 | 95001 |     1 |   NULL |     NULL |
|    2689 | 24001 |    42 |      3 |        1 |
|    2689 | 99001 |     7 |     10 |        1 |
|    2689 | 27001 |     7 |   NULL |     NULL |
|    2755 | 27001 |     2 |   NULL |     NULL |
|    2689 | 98002 |     3 |     10 |        1 |
|    2755 | 98002 |     2 |     10 |        3 |
+---------+-------+-------+--------+----------+
16 rows in set (0.0051 sec)
```

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

SELECT p.FName, p.LName, m.Ground, m.Date, b.NRuns FROM Player p INNER JOIN Batting b ON p.PlayerID = b.PID INNER JOIN ODIMatch m ON b.MatchID = m.MatchID WHERE m.Team1 = 'India' OR m.Team2 = 'India';

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT p.FName, p.LName, m.Ground, m.Date, b.NRuns
                                                -> FROM Player p
                                                -> INNER JOIN Batting b ON p.PlayerID = b.PID
                                                -> INNER JOIN ODIMatch m ON b.MatchID = m.MatchID
                                                -> WHERE m.Team1 = 'India' OR m.Team2 = 'India';
+-----------+-------------+----------+------------+-------+
| FName     | LName       | Ground   | Date       | NRuns |
+-----------+-------------+----------+------------+-------+
| Yuvraj    | Singh       | Colombo  | 2008-08-24 |     0 |
| MS        | Dhoni       | Colombo  | 2008-08-24 |    71 |
| Sanath    | Jayasuriya  | Colombo  | 2008-08-24 |    60 |
| Chaminda  | Vaas        | Colombo  | 2008-08-24 |    17 |
| Muthiah   | Murlitharan | Colombo  | 2008-08-24 |     1 |
| Sachin    | Tendulkar   | Brisbane | 2008-03-04 |    91 |
| Yuvraj    | Singh       | Brisbane | 2008-03-04 |    38 |
| MS        | Dhoni       | Brisbane | 2008-03-04 |    36 |
| Adam      | Gilchrist   | Brisbane | 2008-03-04 |     2 |
| Ricky     | Ponting     | Brisbane | 2008-03-04 |     1 |
| Andrew    | Symonds     | Brisbane | 2008-03-04 |    42 |
| Breet     | Lee         | Brisbane | 2008-03-04 |     7 |
| Praveen   | Kumar       | Brisbane | 2008-03-04 |     7 |
| Praveen   | Kumar       | Colombo  | 2008-08-24 |     2 |
| Harbhajan | Singh       | Brisbane | 2008-03-04 |     3 |
| Harbhajan | Singh       | Colombo  | 2008-08-24 |     2 |
+-----------+-------------+----------+------------+-------+
16 rows in set (0.0019 sec)
```

7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

SELECT CONCAT(p.FName, ' ', p.LName) AS PlayerName, m.Ground, m.Date, b.NRuns, bw.Nwickets FROM Player p INNER JOIN Batting b ON p.PlayerID = b.PID INNER JOIN Bowling bw ON b.PID = bw.PID AND b.MatchID = bw.MatchID INNER JOIN ODIMatch m ON b.MatchID = m.MatchID;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT CONCAT(p.FName, ' ', p.LName) AS PlayerName,
                                                ->        m.Ground, m.Date, b.NRuns, bw.Nwickets
                                                -> FROM Player p
                                                -> INNER JOIN Batting b ON p.PlayerID = b.PID
                                                -> INNER JOIN Bowling bw ON b.PID = bw.PID AND b.MatchID = bw.MatchID
                                                -> INNER JOIN ODIMatch m ON b.MatchID = m.MatchID;
+---------------------+----------+------------+-------+-----------+
| PlayerName          | Ground   | Date       | NRuns | Nwickets |
+---------------------+----------+------------+-------+-----------+
| Breet Lee           | Brisbane | 2008-03-04 |     7 |         1 |
| Andrew Symonds      | Brisbane | 2008-03-04 |    42 |         1 |
| Yuvraj Singh        | Brisbane | 2008-03-04 |    38 |         0 |
| Chaminda Vaas       | Colombo  | 2008-08-24 |    17 |         1 |
| Muthiah Murlitharan | Colombo  | 2008-08-24 |     1 |         1 |
| Sanath Jayasuriya   | Colombo  | 2008-08-24 |    60 |         0 |
| Yuvraj Singh        | Colombo  | 2008-08-24 |     0 |         2 |
| Harbhajan Singh     | Colombo  | 2008-08-24 |     2 |         3 |
| Harbhajan Singh     | Brisbane | 2008-03-04 |     3 |         1 |
+---------------------+----------+------------+-------+-----------+
9 rows in set (0.0016 sec)
```

8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

SELECT CONCAT(p1.FName, ' ', p1.LName) AS Player1, CONCAT(p2.FName, ' ', p2.LName) AS Player2, p1.Bplace FROM Player p1 INNER JOIN Player p2 ON p1.Bplace = p2.Bplace AND p1.PlayerID < p2.PlayerID;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT CONCAT(p1.FName, ' ', p1.LName) AS Player1,
                                             ->        CONCAT(p2.FName, ' ', p2.LName) AS Player2,
                                             ->        p1.Bplace
                                             -> FROM Player p1
                                             -> INNER JOIN Player p2 ON p1.Bplace = p2.Bplace
                                             -> AND p1.PlayerID < p2.PlayerID;
+-----------------+------------------+-----------+
| Player1         | Player2          | Bplace    |
+-----------------+------------------+-----------+
| Virender Sehwag | Ishant Sharma    | Delhi     |
| Gary Kirsten    | Jacques Kallis   | Cape Town |
| Gary Kirsten    | Herschelle Gibbs | Cape Town |
| Jacques Kallis  | Herschelle Gibbs | Cape Town |
+-----------------+------------------+-----------+
4 rows in set (0.0056 sec)
```

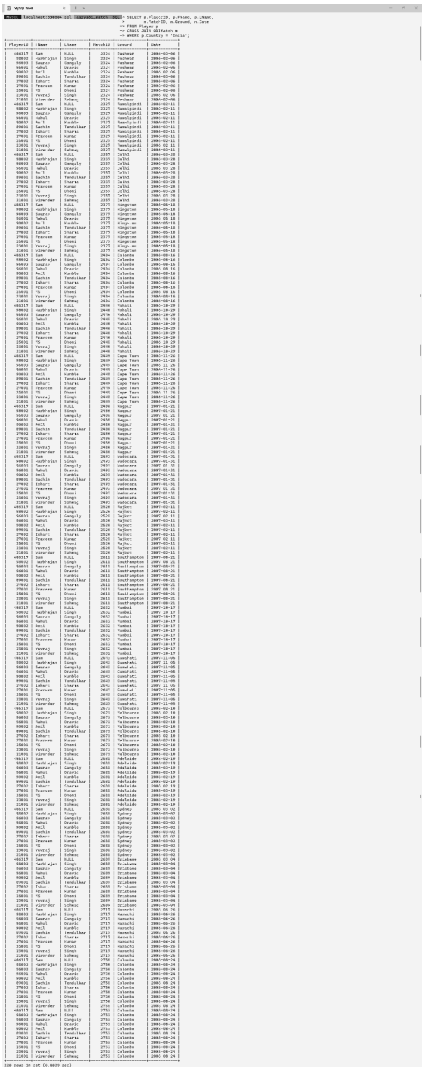
9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

SELECT m1.MatchID AS Match1, m2.MatchID AS Match2, m1.Ground, m1.Date AS Date1, m2.Date AS Date2 FROM ODIMatch m1 INNER JOIN ODIMatch m2 ON m1.Ground = m2.Ground AND m1.MatchID < m2.MatchID AND m1.Date <> m2.Date;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT m1.MatchID AS Match1, m2.MatchID AS Match2,
                                             ->        m1.Ground, m1.Date AS Date1, m2.Date AS Date2
                                             -> FROM ODIMatch m1
                                             -> INNER JOIN ODIMatch m2 ON m1.Ground = m2.Ground
                                             -> AND m1.MatchID < m2.MatchID
                                             -> AND m1.Date <> m2.Date;
+--------+--------+---------+------------+------------+
| Match1 | Match2 | Ground  | Date1      | Date2      |
+--------+--------+---------+------------+------------+
|   2404 |   2750 | Colombo | 2006-08-16 | 2008-08-24 |
|   2404 |   2755 | Colombo | 2006-08-16 | 2008-08-24 |
+--------+--------+---------+------------+------------+
2 rows in set (0.0019 sec)
```

10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

SELECT p.PlayerID, p.FName, p.LName, m.MatchID, m.Ground, m.Date FROM Player p CROSS JOIN ODIMatch m WHERE p.Country = 'India';



11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

SELECT b.PID, SUM(b.NRuns) AS TotalRuns FROM Batting b INNER JOIN ODIMatch m ON b.MatchID = m.MatchID WHERE m.Ground = 'Colombo' GROUP BY b.PID;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT b.PID, SUM(b.NRuns) AS TotalRuns
                                                 -> FROM Batting b
                                                 -> INNER JOIN ODIMatch m ON b.MatchID = m.MatchID
                                                 -> WHERE m.Ground = 'Colombo'
                                                 -> GROUP BY b.PID;

+-------+------------+
| PID   | TotalRuns |
+-------+------------+
| 23001 |          0 |
| 25001 |         71 |
| 91001 |         60 |
| 94002 |         17 |
| 92002 |          1 |
| 27001 |          2 |
| 98002 |          2 |
+-------+------------+
7 rows in set (0.0058 sec)
```

12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

SELECT p.PlayerID, p.FName, p.LName, SUM(b.NRuns) AS TotalCareerRuns FROM Player p INNER JOIN Batting b ON p.PlayerID = b.PID GROUP BY p.PlayerID, p.FName, p.LName ORDER BY TotalCareerRuns DESC;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT p.PlayerID, p.FName, p.LName,
                                                 ->        SUM(b.NRuns) AS TotalCareerRuns
                                                 -> FROM Player p
                                                 -> INNER JOIN Batting b ON p.PlayerID = b.PID
                                                 -> GROUP BY p.PlayerID, p.FName, p.LName
                                                 -> ORDER BY TotalCareerRuns DESC;
+----------+-----------+-------------+-----------------+
| PlayerID | FName     | LName       | TotalCareerRuns |
+----------+-----------+-------------+-----------------+
|    25001 | MS        | Dhoni       |             107 |
|    89001 | Sachin    | Tendulkar   |              91 |
|    91001 | Sanath    | Jayasuriya  |              60 |
|    24001 | Andrew    | Symonds     |              42 |
|    23001 | Yuvraj    | Singh       |              38 |
|    94002 | Chaminda  | Vaas        |              17 |
|    27001 | Praveen   | Kumar       |               9 |
|    99001 | Breet     | Lee         |               7 |
|    98002 | Harbhajan | Singh       |               5 |
|    99002 | Adam      | Gilchrist   |               2 |
|    92002 | Muthiah   | Murlitharan |               1 |
|    95001 | Ricky     | Ponting     |               1 |
+----------+-----------+-------------+-----------------+
12 rows in set (0.0046 sec)
```

13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

SELECT p.PlayerID, p.FName, p.LName, p.Country FROM Player p LEFT JOIN Bowling bw ON p.PlayerID = bw.PID WHERE bw.PID IS NULL;

```
MySQL  localhost:33060+ ssl  aarya61_match  SQL > SELECT p.PlayerID, p.FName, p.LName, p.Country
                                                 -> FROM Player p
                                                 -> LEFT JOIN Bowling bw ON p.PlayerID = bw.PID
                                                 -> WHERE bw.PID IS NULL;
+----------+------------+-----------+--------------+
| PlayerID | FName      | LName     | Country      |
+----------+------------+-----------+--------------+
|    21001 | Virender   | Sehwag    | India        |
|    25001 | MS         | Dhoni     | India        |
|    27001 | Praveen    | Kumar     | India        |
|    27002 | Ishant     | Sharma    | India        |
|    89001 | Sachin     | Tendulkar | India        |
|    90001 | Brain      | Lara      | West Indies  |
|    90002 | Anil       | Kumble    | India        |
|    92001 | Shane      | Warne     | Australia    |
|    92003 | Inzamam    | UI-Huq    | Pakistan     |
|    93002 | Heath      | Streak    | Zimbabwe     |
|    93003 | Gary       | Kirsten   | South Africa |
|    94004 | Stephen    | Fleming   | New Zealand  |
|    95001 | Ricky      | Ponting   | Australia    |
|    95002 | Shaun      | Pollock   | South Africa |
|    95003 | Jacques    | Kallis    | South Africa |
|    96001 | Rahul      | Dravid    | India        |
|    96002 | Herschelle | Gibbs     | South Africa |
|    96003 | Saurav     | Ganguly   | India        |
|    97004 | Daniel     | Vettori   | New Zealand  |
|    98001 | Shahid     | Afridi    | Pakistan     |
|    99002 | Adam       | Gilchrist | Australia    |
|    99003 | Michael    | Vaughan   | England      |
|   300311 | Michael    | Clark     | NULL         |
|   406317 | Sam        | NULL      | India        |
+----------+------------+-----------+--------------+
24 rows in set (0.0012 sec)
```